

Chapter 3

3.2

(a) As x increase 1 unit, the percentage of times the starting pitcher pitched a complete game will decrease 6.62%.

(b) In the linear prediction, when $x = 12$, the estimated percentage becomes -0.1014 (-10.14%), which is below zero. In the logistic regression, when $x = 12$, the estimated percentage is 0.034 (3.4%). The prediction of logistic regression is more plausible since it is between 0 and 1, while the prediction of linear model is below 0, which isn't reasonable.

3.5

(a) The estimated linear regression model is: $\hat{P}(Y = 1) = -0.145 + 0.323 \times \text{weight}$.

When weight is equal to 0, the estimated probability of $Y = 1$ is -0.145 . As weight increase 1 unit, the estimated probability will increase by 0.323.

At a weight of 5.20 kg, the estimated probability is 1.533. Since 1.533 is larger than 1, it's not reasonable.

(b) The estimated logistic regression is: $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = -3.695 + 1.815 \times \text{weight}$.

At a weight of 5.20 kg, $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 5.744 \Rightarrow \hat{P}(Y = 1) = \frac{e^{5.744}}{1+e^{5.744}} = 0.9968$.

3.6

(a) The estimated logistic regression is: $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 3.1870 - 0.5901x$. Since the coefficient of x is negative, it means that as x increases, the probability of being a Democrat will decrease.

(b) The 95% Wald CI for the effect of political ideology is $\hat{\beta} \pm 1.96(SE) = -0.5901 \pm 1.96(0.1564) = (-0.897, -0.284)$. Compared to the profile CI $(-0.916, -0.298)$, Wald CI is shorter.

(c) The Wald test statistic is $z = \frac{\hat{\beta}}{SE} = -\frac{0.5901}{0.1564} = -3.772$, the p-value is equal to 0.000162, which implies that the effect of political ideology is significant.

(d) The likelihood-ratio test statistic is $24.7983 - 7.7894 = 17.0089$ which has an approximate χ_1^2 , the p-value is equal to 0.0000372, which implies that the effect of political ideology is significant.

(e) The number of Fisher scoring iterations is 4, which means that the ML estimates of the parameters done by Fisher scoring required only 4 iterations to converge.

3.8

The estimated logistic regression model is: $\text{logit} [\hat{P}(Y = 1)] = \log \left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)} \right) = -3.866 + 0.397x$.

The Wald test for effect of snoring: $z = \frac{0.397}{0.05} = 7.945 > z_{0.975} = 1.96$, which implies the snoring effect is significant.

The Wald 95% CI of the effect of snoring is $\hat{\beta} \pm 1.96(SE) = 0.397 \pm 1.96(0.05) = (0.299, 0.495)$.

3.11

The estimated Poisson regression model is: $\log(\hat{\mu}) = 1.6094 + 0.5878x$.

$e^{0.5878} = 1.8$, which implies that treatment B has an 80.0% increase in the estimated mean number of imperfections.

3.12

(a) Hypothesis test: $\begin{cases} H_0: \beta = 0 \\ H_1: \beta \neq 0 \end{cases}$.

Wald test: $z = \frac{0.5878}{0.1764} = 3.332 > z_{0.975} = 1.96$.

Likelihood-ratio test: $2(L_1 - L_0) = 27.857 - 16.268 = 11.589 > \chi^2_{(1),0.95} = 3.84$.

Since the Wald test and likelihood-ratio test are both significant, it implies treatment A is significantly better than treatment B.

(b)

The Wald 95% CI for β is $\hat{\beta} \pm 1.96(SE) = 0.5878 \pm 1.96(0.1764) = (0.242, 0.934)$. The Wald 95% CI for μ_B/μ_A is $(e^{0.242}, e^{0.934}) = (1.274, 2.544)$.

The profile likelihood 95% CI for β is $(0.2469, 0.9401)$. The profile likelihood 95% CI is $(e^{0.4597}, e^{0.7145}) = (1.28, 2.56)$.

3.13

(a) $\log(\mu) = -0.428 + 0.5893x$. When $x = 2.44$, the estimated mean response is 2.744.

(b) $\hat{\beta} = 0.5893$, $e^{0.589} = 1.803$, which means a 1-cm increase in the weight has an 80.3% increase in the estimated mean number of satellites.

The Wald 95% CI for β is $\hat{\beta} \pm 1.96(SE) = 0.5893 \pm 1.96(0.065) = (0.4619, 0.7167)$. the Wald 95% CI for the multiplicative effect of a 1-kg increase is $(e^{0.4619}, e^{0.7167}) = (1.587, 2.048)$.

The profile likelihood 95% CI for β is $(0.4597, 0.7145)$. The profile likelihood 95% CI for the multiplicative

effect of a 1-kg increase is $(e^{0.4597}, e^{0.7145}) = (1.584, 2.043)$.

(c)

The Wald test statistic is $z = \frac{\hat{\beta}}{SE} = \frac{0.5893}{0.065} = 9.064 > z_{0.975} = 1.96$.

The likelihood-ratio test is $2(L_1 - L_0) = 632.79 - 560.87 = 71.92 > \chi^2_{(1),0.95} = 3.84$.

Since the Wald test and likelihood-ratio test are both significant, it implies the effect of weight is significant.

3.15

(a) True. (b) False, the variance of Y does not need to be constant at all values of explanatory variables.